Mapping a Decade of Research at IIT Kharagpur and IIT Roorkee (2016–2026)

Assignment 2, Dynamical Processes in Complex Networks

Team 3
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***Abstract*—We collected the papers of IIT Kharagpur and IIT Roorkee from 2016 to 2026 and linked them to 1,638 faculty. We study them at the institution, department and faculty level, and we build a citation network and a co-authorship network. Both IITs doubled their output, their citations are heavy-tailed, and their faculty work in tight groups inside their own IIT.**

I. Introduction

We look at research at three levels, the **institution**, its **departments** and its **faculty**. Counts of papers and citations describe each level, but they do not show how the research is *connected*. Networks show this. A **citation network** shows how ideas move between papers. A **co-authorship network** shows how people work together.

This report follows that order. It goes from the institution to the faculty, then to the two networks, and then to the links with other Indian institutes. Table I gives the main numbers.

TABLE I
The dataset at a glance

	IIT Kharagpur	IIT Roorkee
Papers, 2016–2026	30,019	22,844
Papers in 2016 → 2025	1,782 → 3,429	1,335 → 2,932
Journal papers	21,988	16,970
Q1 share of ranked journal papers	64.4%	59.5%
CORE A* / A conference papers	118 / 97	11 / 37
Faculty with at least one paper	934	704
Median faculty h-index	7	7

II. Data and Method

Collecting the papers. We scraped publication records from public sources. These are Crossref, Europe PMC, DOAJ, INSPIRE, ORCID, DBLP and Semantic Scholar. We searched in two ways. The first way uses the institute name in the author affiliations. The second way uses the faculty list of each IIT. This gave 33,393 records for KGP and 25,775 records for Roorkee.

Filling in missing values. Many records had missing fields. We used **OpenAlex** to get the DOI, type, venue, ISSN, citations, references, author affiliations and research topic. We used **Scimago** to get the journal quartile, by ISSN and publication year. We used **CORE** to get the conference rank. We did not guess any value. If we could not find a value, we left it blank.

Checking the affiliation. We kept a paper only if the IIT is in the author affiliations on the paper. This removes namesakes and papers from before a person joined the IIT. After this check we have **30,019** papers for KGP and **22,844** papers for Roorkee.

Linking papers to faculty. The brief asks for one row for each paper and faculty member. We linked papers to faculty with five rules, from the most reliable to the least reliable. These are the **ORCID**, the faculty profile where we found the paper, the exact **name**, the **name and department**, and the OpenAlex author ID. Extra checks stop wrong matches, for example two faculty with the same common surname. About **85%** of the papers have at least one faculty member (84.7% for KGP and 84.4% for Roorkee). We credited the other papers to the institute.

Deliverables. The four files have 63,122 rows (`group3_data`), 544,794 citation edges, 160,214 author pairs and 5,847 Indian partners.

Limits. We include all fields, which are science, engineering, health and social sciences. 2026 is a **partial year**, and the citation counts are from October 2026. The h-index uses only the 2016–2026 papers in our dataset. Thus it is lower than a Google Scholar h-index.

III. Institution-Level Statistics

A. Publication trend

The simplest measure of research activity is the **number of papers per year**. We look at it first because it sets the scale for all other results. More papers give more citations and larger networks. Thus we must keep the size of each IIT in mind when we compare them.

Both IITs grew steadily (Fig. 1). KGP went from 1,782 papers in 2016 to 3,429 papers in 2025, which is **1.9 times** more. Its peak was 3,580 papers in 2024. Roorkee went from 1,335 papers to 2,932 papers, which is **2.2 times** more. KGP is larger every year, but Roorkee grows faster. Not all paper types grew at the same rate. At KGP, conference papers grew 2.4 times and journal papers grew 1.7 times. At Roorkee the values are 2.9 and 2.0. Preprints went from almost zero to 150 to 200 each year. Thus part of the growth is a change in *how* the IITs publish, and not only in how much. The drop in 2026 is because the year is not complete.

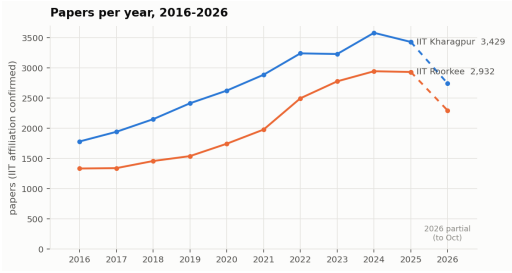


Fig. 1. Papers per year. The dashed part is 2026, which is a partial year.

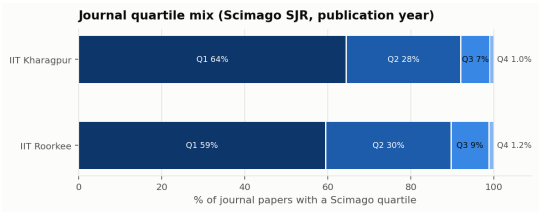


Fig. 2. Journal quartile mix (Scimago, year of publication).

B. Journal quality

A paper count does not show quality. The **Scimago quartile** ranks each journal in its subject by its SJR score. Then it divides the list into four equal parts. Q1 is the top 25%. Scimago ranks journals *inside* each subject. Thus the Q1 share lets us compare quality across fields that cite at very different rates.

About 94% of the journal papers of both IITs have a quartile. Of these papers, **64.4%** at KGP and **59.5%** at Roorkee are in Q1 (Fig. 2). Between 28% and 30% are in Q2, and only about 1% are in Q4. We expect this result from two of the oldest IITs, which publish mostly in established journals.

For conferences we use the **CORE** rank. CORE ranks mostly computing conferences. Here KGP is far ahead, with 118 A* papers and 97 A papers. Roorkee has 11 and 37. This result makes sense. Computer Science is the largest department of KGP by papers (Table III), and it gives 12.2% of KGP papers. At Roorkee, Computer Science gives 8.0% of papers.

C. Research areas and journals

Next we ask *what* each IIT works on. For this we use the OpenAlex **field** of each paper. The research areas are important for the rest of the report. They control which papers can cite each other and which faculty can work together. In Section VII we use them again to compare the IITs with their partners.

Engineering is the largest field at both IITs. It has 31.6% of KGP papers and 38.6% of Roorkee papers. After Engineering, the two IITs are different. The second field at KGP is **Computer Science** (12.2%). The second field at Roorkee is **Environmental Science** (11.3%). This agrees with the large Civil Engineering department of Roorkee, which is its largest department (Table III), and with its departments that study water. Materials Science is third at both IITs, with about 9%. The top journals of KGP are Physics of Fluids (330 papers), Scientific Reports (194) and ACS Applied Materials and Interfaces (152). The top journals of Roorkee are Scientific Reports (133), the International Journal of Biological Macromolecules (132) and Physical Review C (117). All of them are Q1 journals. Some are broad journals, and others match a strong area of each IIT, such as fluid mechanics at KGP and nuclear physics at Roorkee.

TABLE II
MOST-CITED PAPERS (WORLDWIDE, OCTOBER 2026)

	Paper (year)	**Cit.**
KGP	Guidelines for the use and interpretation of assays for monitoring autophagy (2016)	4,434
KGP	Soil carbon 4 per mille (2017)	2,020
KGP	SymPy, symbolic computing in Python (2017)	1,502
Roork.	The second data release from the European Pulsar Timing Array (2023)	1,069
Roork.	A comprehensive review on NSGA-II for multi-objective combinatorial optimization (2021)	971
Roork.	Microwave–material interaction phenomena, heating mechanisms ... (2016)	876

D. Most-cited papers

The last measure at this level is the number of **worldwide citations** of each paper. These papers are the most visible work of each IIT. But some of them tell us more about the type of paper than about the institute.

Table II shows the top three papers of each IIT. The top papers of KGP are of three types. These are **large collaborations** (the autophagy guidelines, with 4,434 citations), **software tools** (SymPy and iFogSim) and **reviews**. At Roorkee, at least five of the top ten papers are **reviews or surveys**. Their topics are NSGA-II, differential evolution, metaheuristics, reinforcement learning and graphene composites. The top paper of Roorkee is also a large collaboration, the European Pulsar Timing Array. This result makes sense. Everyone who uses a topic or a software tool cites its review or its tool paper. Thus these papers get citations much faster than a typical research paper.

IV. FACULTY AND DEPARTMENT LEVEL

The institution numbers are sums over people. Now we look at the people, and then at the departments that group them.

A. h-index of faculty

A faculty member has an **h-index** of h if h of their papers have at least h citations each. This measure is useful here because it combines *how much* a person publishes and *how much* other people cite it. One very large paper, such as the large collaborations above, cannot increase it alone.

The median h-index is **7** at both IITs. The mean is 9.2 at KGP and 8.7 at Roorkee. The distribution is strongly right-skewed (Fig. 3). Many faculty have a small h-index, and a few have a very large one. The highest value is 63 at KGP

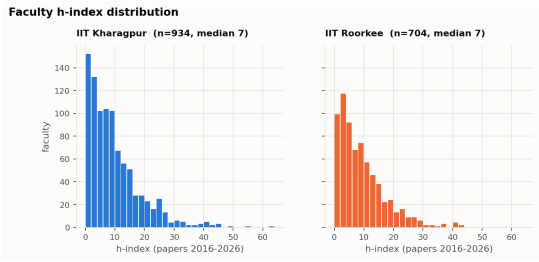


Fig. 3. Distribution of the faculty h-index (2016–2026 papers in our dataset).

TABLE III
TOP 5 DEPARTMENTS OF EACH IIT BY NUMBER OF PAPERS

Department	Fac.	Papers	Cit.	Q1
IIT Kharagpur				
Computer Sci. & Engg.	52	2,836	30,227	67.8%
Mechanical Engg.	54	1,998	29,849	70.8%
Civil Engg.	45	1,796	42,007	70.5%
Chemistry	42	1,712	37,324	72.9%
Agricultural & Food Engg.	53	1,632	35,235	61.2%
IIT Roorkee				
Civil Engg.	75	2,188	30,002	57.0%
Electronics & Comm. Engg.	45	1,887	16,262	49.6%
Mech. & Industrial Engg.	54	1,745	31,497	61.2%
Electrical Engg.	48	1,738	17,169	68.2%
Physics	44	1,545	21,154	65.3%

(Narayan Chandra Das, Rubber Technology), and the next is 54 (Makarand M. Ghangrekar, Civil Engineering). At Roorkee the highest value is 42 (Zillur Rahman, Management Studies, and Soumitra Satapathi, Physics). The low median makes sense. We count only ten years of papers, and recent papers had little time to get citations.

B. Departments

Departments are between the person and the institute. They are the natural unit to compare research areas inside one IIT. Table III shows the five departments with the most papers at each IIT.

At KGP, Computer Science has the most papers, but Civil Engineering has the most citations (42,007). Civil Engineering includes environmental engineering, and two of the five highest h-index values at KGP are in this department. At Roorkee, Civil Engineering has the most papers and the most faculty (75). The Q1 share is similar in the engineering departments, between 50% and 73%. Mathematics is much lower at both IITs, at about 44%. We expect this, because mathematics journals have lower impact scores in general. Small departments can also be strong. Rubber Technology at KGP has only 9 faculty, but its median h-index is 22.

V. CITATION NETWORK

Up to now, all citations had the same value. A network lets us ask *who* cites whom.

A. Construction

Each **node** is a paper of our two IITs. A **directed edge** $A \to B$ means that paper A cites paper B, and both papers are in our dataset. The edges have a direction because a citation goes one way, from a newer paper to an older paper. The network has **52,584** papers and **96,030** edges. KGP has 29,997 papers and 53,798 edges, and Roorkee has 22,804 papers and 38,648 edges. The network is very sparse. The density is 3.5×10^{-5} and the mean in-degree is 1.8. About **half** of the papers get no citation from another paper of the two IITs. The largest weakly connected component has 53% of the papers. We expect this, because the papers are in many fields that do not cite each other.

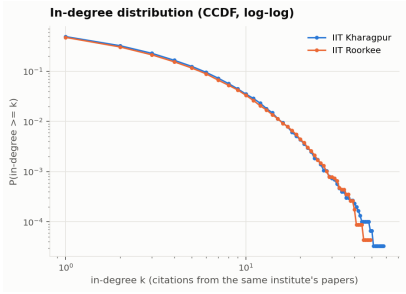


Fig. 4. In-degree distribution of the citation network (CCDF, log-log).

B. In-degree distribution

As we know from our lectures already, the in-degree of a node is the number of edges that point to it. Here it is the number of papers in our dataset that cite a paper. Its distribution shows if attention is spread evenly or goes to a small number of papers.

Fig. 4 shows $P(\text{in-degree} \geq k)$ on a log-log scale. The curve decreases slowly over two orders of magnitude. Thus the distribution is **heavy-tailed**. Most papers get zero, one or two internal citations, but a few papers get 40 to 58. This is the "rich get richer" effect, or preferential attachment, from our lectures. People find a cited paper more easily, so they cite it again. The curves of the two IITs are almost the same. This tells us that the effect comes from how citation works, and not from one institute.

C. PageRank

As we know from our lectures already, **PageRank** gives a score to a node. The score depends on how many nodes point to it *and* on how important those nodes are.

$$PR(i) = \frac{1-\alpha}{N} + \alpha \sum_{j \to i} \frac{PR(j)}{k_{\text{out}}(j)}, \qquad \alpha = 0.85$$

Here N is the number of papers and $k_{\text{out}}(j)$ is the number of papers that j cites. The in-degree gives the same value to every citation. PageRank gives more value to a citation from an influential paper. Thus it finds papers that are the *base* of other well-cited work.

Our data shows this clearly (Table IV). The top paper by PageRank is a 2016 KGP paper on MoS_2 heterojunctions. It has only 20 internal citations and its in-degree rank is 209. But the papers that cite it have many citations themselves.

TABLE IV
TOP 5 PAPERS BY PAGERANK (BOTH IITS), WITH IN-DEGREE AND IN-DEGREE RANK

	Paper (year)	In	Rank
KGP	Hydrothermal growth of few layer 2H-MoS_2 for heterojunction ... (2016)	20	209
KGP	Uniform electric-field-induced lateral migration of a sedimenting ... (2016)	58	1
KGP	V_2O_5 microflower decorated cathode for enhancing power ... (2016)	23	117
Roork.	A review of power electronic converters for variable speed ... (2018)	49	4
Roork.	Synergistic dynamics of nitrogen and phosphorous ... (2016)	21	174

There is also an **age bias**. Seven of the top ten papers are from 2016, and the Spearman correlation between PageRank and publication year is -0.39. This makes sense. In a citation network the score moves almost only from newer papers to older papers. Thus the oldest papers in our window collect the most score.

D. Citation flow with other Indian institutes

The network above stays inside our two IITs. To see their place in Indian research, we also count citations to and from other Indian institutes. The **in-flow** is the number of citations from papers of other Indian institutes to our papers. The **out-flow** is the number of citations from our papers to their papers. The **impact ratio** is the in-flow divided by the out-flow. A ratio above 1 means that Indian research cites the IIT more than the IIT cites Indian research.

KGP has an in-flow of 114,443 and an out-flow of 129,758, so its ratio is **0.88**. Roorkee has 96,963 and 117,255, so its ratio is **0.83**. At first, it seems that both IITs give more citations than they get. Fig. 5 shows the reason. In 2016 our papers already cited thousands of older Indian papers (the KGP out-flow was 5,764). But almost no paper had cited them yet (the KGP in-flow was 217), because citations take years to arrive. The in-flow grows much faster over time. Since **2023** it is larger than the out-flow at both IITs. Thus the total ratio below 1 comes mostly from the 2016–2026 window.

The largest Indian source of citations to KGP is IIT Roorkee (3,105 citations). VIT (3,022) and AcSIR (2,885) are next. For Roorkee the largest sources are VIT (2,565), UPES (2,533) and AcSIR (2,211). Large universities such as VIT and Saveetha cite our IITs much more than our IITs cite them (the KGP ratio with Saveetha is 4.7).

VI. CO-AUTHORSHIP NETWORK

The citation network shows how *ideas* move. The co-authorship network shows how *people* work together. This is closer to how research groups really work.

A. Construction

Each **node** is a faculty member of one of the two IITs. An **undirected edge** connects two faculty who wrote at least one paper together. The edge weight is the number of joint papers. The network has **1,638** faculty and **3,329** edges, and its density is 0.0025. On average, a faculty member has 4.1 faculty co-authors, and 324 faculty have no faculty co-author. The largest component has 1,260 faculty, which is 77% of all faculty. Fig. 6 shows this component. The two IITs make **two separate clusters**, and only **90** edges connect them. This is natural, because the two campuses are far apart and most work happens inside one campus.

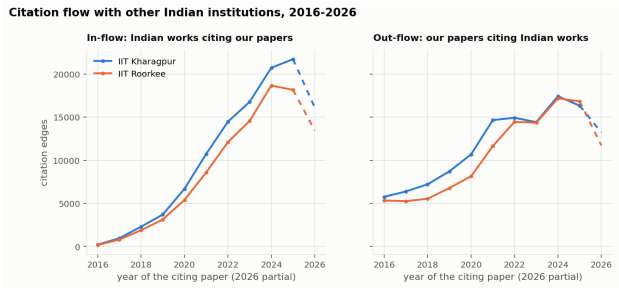


Fig. 5. Citation in-flow and out-flow with other Indian institutes, per year.

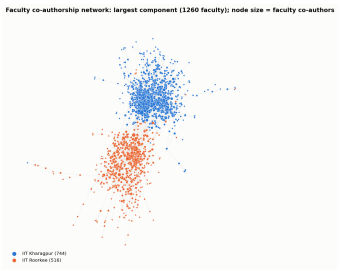


Fig. 6. Largest component of the faculty network. KGP is blue and Roorkee is orange.

B. Degree centrality

As we know from our lectures already, **degree centrality** is the number of neighbours of a node, divided by $n-1$. In this network it is the number of faculty colleagues that a person wrote papers with. Thus it finds the people who collaborate the most.

The highest values are for Santanu Dhara (KGP, Medical Science and Technology, 52 faculty co-authors), Suman Chakraborty (KGP, Mechanical Engineering, 49) and Partha Roy (Roorkee, Biosciences and Bioengineering, 44). Medical science and biosciences are interdisciplinary areas. A project in these areas needs people from many departments, so we expect a high degree here.

C. Betweenness centrality

As we know from our lectures already, **betweenness centrality** measures how often a node is on the shortest path between two other nodes. A person with a high betweenness is a **bridge**. Without this person, some groups are far apart or not connected. This role is different from having many co-authors.

The top bridges are Partha Roy at Roorkee (0.048), and Suman Chakraborty (0.030) and Pabitra Mitra (Computer Science, 0.029) at KGP. Pabitra Mitra shows the difference between the two measures. He has fewer co-authors than the top people by degree (35), but his co-authors are in many different groups. Many departments use computer science methods. Thus it makes sense that a computer scientist connects them.

D. Communities

A **community** is a group of nodes with many edges inside the group and few edges to other groups. We use the **Louvain** algorithm, which finds the groups with the highest **modularity**.

$$Q = \frac{1}{2m} \sum_{i,j} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j)$$

The modularity compares the number of edges inside the groups with the number we expect at random. Here the communities are the real research groups. These groups can follow department lines, or not.

We get $Q = 0.76$ and **27 communities** with 5 or more faculty. In 23 of these 27 communities, 90% or more of the members are from one IIT. This again shows that collaboration stays inside each campus. The communities are **interdisciplinary**. In a typical community, the largest department has only about 44% of the members. For example, the largest community has 146 faculty from Roorkee. Biosciences is its largest department, but its members come from 20 departments. The result is stable. Over 10 random seeds, the mean Adjusted Rand Index is 0.83.

E. Comparison with random networks

As we know from our lectures already, the **clustering coefficient** is the probability that two co-authors of a person also wrote a paper together. A value such as 0.17 has a meaning only when we compare it with a baseline. Thus we compare the network with two random models of the same size, with 10 samples each (Table V). The first model is an **Erdős–Rényi** (ER) graph with the same number of edges. The second model is a **degree-preserving** rewiring that keeps the degree of every node.

The real network has about **90 times** more clustering than the ER graph and 17 times more than the rewired graph. But its mean shortest path is almost the same as in the ER graph. This is the **small-world** property. The network has tight local groups, but the paths between people are still short. This makes sense for research groups. A new project often brings in a colleague of a colleague. The modularity is also much higher than in the two random models. Thus the communities above are real, and the algorithm did not create them. The positive assortativity (0.19) shows that well-connected faculty usually work with other well-connected faculty.

VII. Connections with Other Indian Institutes

The faculty network stays inside the two IITs. As a last step, we look at co-authorship with all other Indian institutes.

TABLE V
FACULTY NETWORK AND TWO RANDOM MODELS (MEAN OF 10 SAMPLES)

Metric	Real	ER	Rewired
Clustering coefficient	0.170	0.002	0.010
Modularity (Louvain)	0.766	0.517	0.441
Degree assortativity	0.185	0.00	−0.01
Mean shortest path (giant comp.)	5.28	5.40	4.13

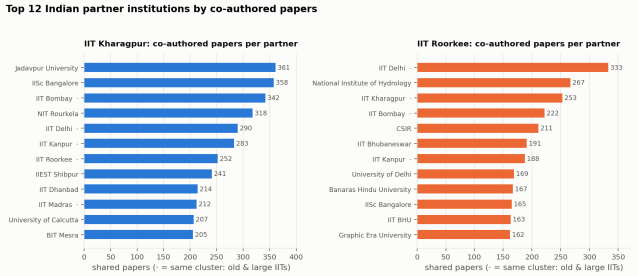


Fig. 7. Top 12 Indian partner institutes of each IIT, by co-authored papers.

A. Partner institutes

We count the **shared papers** between each IIT and each Indian partner. KGP works most with **Jadavpur University** (361 papers), IISc Bangalore (358) and IIT Bombay (342). Roorkee works most with **IIT Delhi** (333), the National Institute of Hydrology (267) and IIT Kharagpur (253) (Fig. 7). These lists make sense by location and by field. The KGP list has several institutes in Kolkata, which are Jadavpur University, IIEST Shibpur and the University of Calcutta. The National Institute of Hydrology is in Roorkee, and it matches the strength of Roorkee in water and environment.

B. Similar or complementary research

Do the IITs work with institutes that do the same research, or with institutes that fill a gap? To measure this, we take the top 20 research subfields of the IIT and of the partner. Then we calculate the **Jaccard similarity**.

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

The value is 1 if the two lists are the same, and 0 if they have nothing in common.

For the top 25 partners of each IIT, the median is **0.60**. Thus most partners do **similar** research. Some strong partners are **complementary**. Examples are the National Institute of Hydrology (0.21, Roorkee) and the Indian Statistical Institute (0.18, KGP). These institutes are specialised. Thus the collaboration is probably on one topic, such as hydrology or statistics.

VIII. Dashboard and Conclusions

All these results are also in an interactive website, with a switch for KGP, Roorkee or both. To conclude, both IITs grew fast and publish mostly in Q1 journals, Indian research cites them more each year, and their faculty work in tight groups inside each campus. Each step is a script in our repository, so anyone can repeat it.